

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1504

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 50

Code of Conduct

I V.Raj kumar S(192525208) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V.Raj kumar

Criteria	Problem Interpretation (2)	Analytical Reasoning (2.5)	Method Selection (2)	Solution Quality (2)	Presentation of Analysis (1.5)	TOTAL (10)
Weightage	20%	25%	20%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

Analytical Problem Solving

1. A cloud datacenter host has 64 CPU cores and supports CPU overcommitment at a ratio of 2.5:1. Each virtual machine (VM) requires 4 vCPUs. Calculate the maximum number of VMs that can be deployed on this host. If actual CPU utilization reaches 85%, analyze whether further VM allocation is advisable.
2. A virtualized storage system provides 20 TB of physical storage with thin provisioning at a ratio of 3:1. Each VM is allocated 500 GB of virtual disk space. Calculate the maximum number of VMs that can be provisioned.
3. Evaluate the security implications of multi-tenancy in cloud datacenters. Analyze how virtualization can both enable and weaken isolation, and recommend mechanisms to strengthen tenant separation.
4. Analyze the role of Service-Oriented Architecture (SOA) in enabling interoperability across heterogeneous cloud platforms. Identify key challenges such as service latency, security, and version control, and propose solutions.
5. A cloud system implementing presence services faces scalability issues during peak usage. Analyze the architectural bottlenecks and recommend design improvements to ensure real-time updates and efficient resource utilization.

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COURSE NAME: CLOUD COMPUTING AND BIG DATA ANALYSIS

Q1. CPU Overcommitment

Given:

- Physical CPU cores = **64**
- CPU overcommitment ratio = **2.5 : 1**
- Each VM requires **4 vCPUs**

Calculation:

Total virtual CPUs available

$$= 64 \times 2.5$$

$$= \mathbf{160 \text{ vCPUs}}$$

Maximum number of VMs

$$= 160 \div 4$$

$$= \mathbf{40 \text{ VMs}}$$

Analysis

If actual CPU utilization reaches **85%**, the host is already under heavy load. Allocating more VMs may increase CPU contention, scheduling delays, and reduce application performance. It can also affect Service Level Agreements (SLAs).

Answer

Maximum number of VMs = 40

Further VM allocation is **not advisable** because CPU utilization is already 85%, leaving very little capacity for additional workloads.

Q2. Thin Provisioning

Given:

- Physical storage = **20 TB**
- Thin provisioning ratio = **3 : 1**

- Each VM virtual disk = **500 GB**

Calculation

Maximum virtual storage

$$= 20 \times 3$$

$$= \mathbf{60\ TB}$$

Convert to GB

$$60\ \text{TB} = \mathbf{60,000\ GB}$$

Maximum number of VMs

$$= 60,000 \div 500$$

$$= \mathbf{120\ VMs}$$

Answer

Maximum number of VMs that can be provisioned = 120

Thin provisioning allows allocating more virtual storage than the available physical storage, but actual storage usage should be monitored to avoid over-allocation.

Q3. Security Implications of Multi-Tenancy

Security Issues

- Multiple tenants share the same physical hardware.
- A compromised VM may attempt to attack neighboring VMs.
- Side-channel attacks may leak sensitive information.
- Misconfigured hypervisors can expose tenant data.
- Resource contention may affect performance and availability.

How Virtualization Enables Isolation

- Each VM has its own operating system.
- Hypervisors isolate CPU, memory, storage, and network resources.
- Virtual networking separates tenant communication.

How Virtualization Weakens Isolation

- Hypervisor vulnerabilities can allow VM escape attacks.
- Shared hardware resources increase attack surface.

- Improper VM migration or snapshots may expose data.

Recommended Mechanisms

- Use secure and updated hypervisors.
- Enable strong authentication and access control.
- Encrypt data at rest and in transit.
- Implement network segmentation using VLANs and virtual firewalls.
- Perform continuous security monitoring and vulnerability assessments.
- Apply regular security patches.

Conclusion

Virtualization provides strong isolation, but proper security controls and monitoring are essential to protect tenants in a multi-tenant cloud environment.

Q4. Role of SOA in Cloud Interoperability

Role of SOA

Service-Oriented Architecture (SOA) enables different cloud platforms and applications to communicate through standardized services. It promotes interoperability, reusability, scalability, and flexibility.

Challenges

1. Service Latency

- Network communication may increase response time.

Solution

- Use caching, load balancing, and optimized service routing.

2. Security

- Unauthorized access and data breaches.

Solution

- Implement authentication, authorization, encryption, and API gateways.

3. Version Control

- Different service versions may create compatibility issues.

Solution

- Use API versioning, backward compatibility, and proper documentation.

Conclusion

SOA improves interoperability among heterogeneous cloud platforms by using standardized services. Addressing latency, security, and version management ensures reliable cloud integration.

Q5. Presence Services Scalability

Architectural Bottlenecks

- Large number of simultaneous users.
- Centralized server becomes overloaded.
- Frequent status updates consume bandwidth.
- Database bottlenecks during peak usage.
- Increased network latency.

Recommended Improvements

- Deploy distributed servers across multiple regions.
- Use load balancers to distribute requests.
- Implement message queues for asynchronous processing.
- Cache frequently accessed presence information.
- Use scalable NoSQL databases.
- Enable auto-scaling of cloud resources.
- Use efficient publish-subscribe messaging protocols.

Benefits

- Improved scalability.
- Reduced latency.
- Better resource utilization.
- High availability.
- Faster real-time presence updates.